

PATH 01 - NEW TERRITORY

Territory Briefing: Torino

Professional province-level preliminary briefing: historical evidence, Atlas map, project-context focus and operational next checks.

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REPORT ID

**ARCUS-P01-
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torino**

VERSION

**ARCUS
Professional
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ANALYSIS

**Territory
Briefing**

**SPATIAL
LEVEL**

**Province
level**

**GENERATED
BY**

**ARCUS
Professional**

01 EXECUTIVE SUMMARY

ARCUS identifies 42 documented precedents in the selected province and translates the Step 3 hazard-overlay state into operational actions for bridge interventions. The dominant territorial signal is **hydraulic**; the leading historical driver is **Hydraulic**.

Recommended use: province-level preliminary screening for bridge interventions, before design, due diligence or site-specific investigation.

FINDING 1

The hydraulic context dominates the territorial reading.

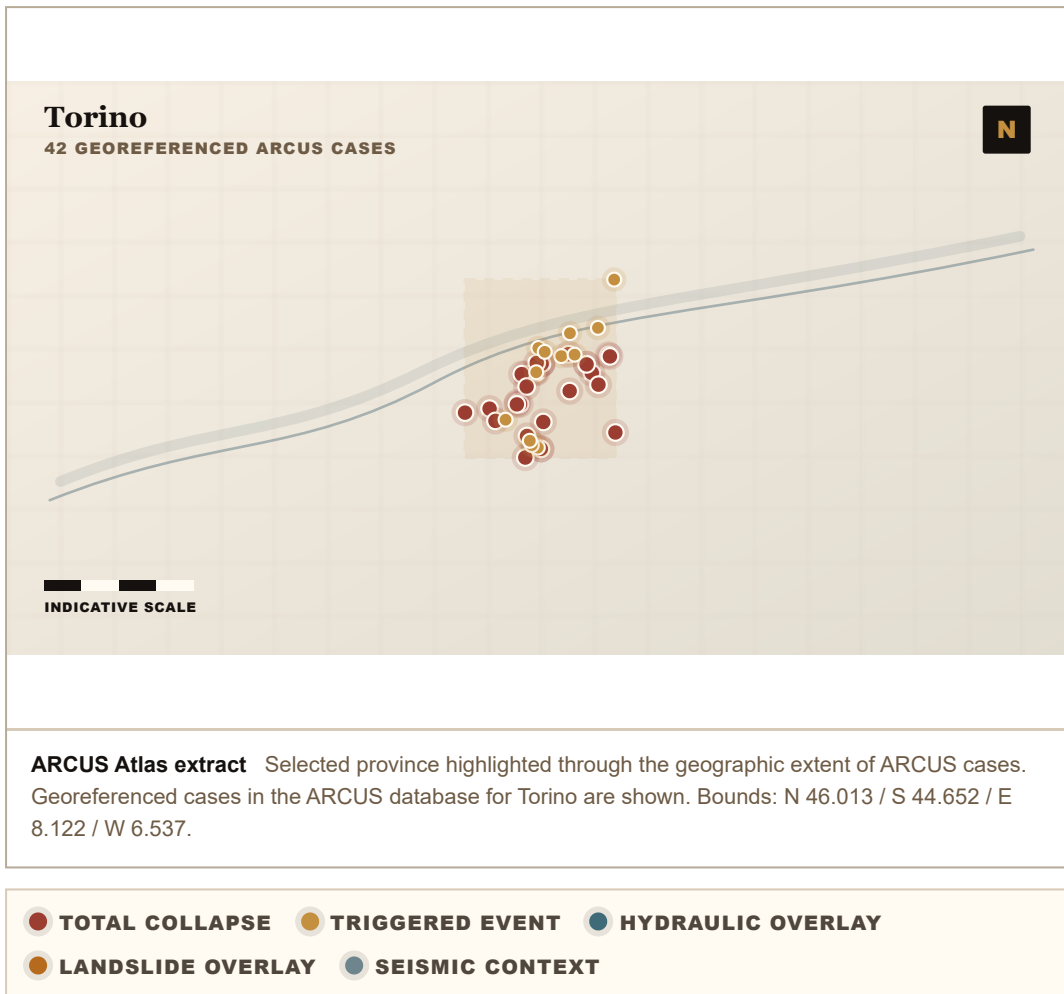
FINDING 2

41 triggered events and 30 total collapses shape the technical priority.

FINDING 3

Very high evidence reliability with 125 linked sources.

02 SELECTED PROVINCE & STEP 3 MAP



Territorial screening map, not design scale. P1/P2/P3 identify the priority precedents referenced in the report.

03 HOW TO READ THIS REPORT

ARCUS combines historical collapse records, public hazard overlays, source-backed evidence and comparable precedents. The objective is to turn overlays and collapse history into operational next steps, not to produce a structural safety certification.

SELECTED PROJECT CONTEXT

bridge

For a bridge project context, the dominant driver (Hydraulic) is translated into preliminary attention points. This section does not prescribe design solutions: it frames checks, data requests and comparison with precedents in the province of Torino.

DOMINANT EXPOSURE

hydraulic / 97

The selected province is mainly driven by hydraulic exposure: 41 matched collapses out of 42 analysed events (98%). Other layers should be read as secondary preliminary-screening context.

04 KEY INDICATORS

PRIORITY INDEX

82 / 100

HIGH ATTENTION

hydraulic

EVIDENCE RELIABILITY

96 / 100

VERY HIGH

125 linked sources

FAILURE PRECEDENT EXPOSURE

72 / 100

HIGH

Hydraulic

HISTORICAL EVENTS

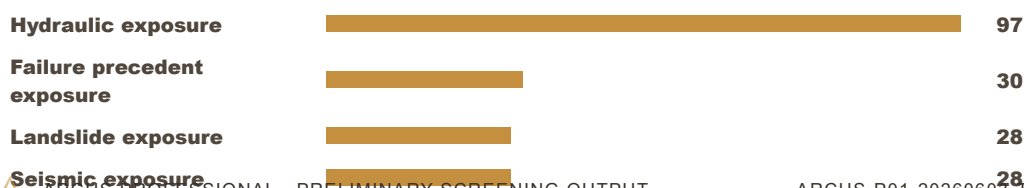
42

98% HYDRAULIC

41 triggered events

05 HAZARD / EXPOSURE CONTEXT

The Step 3 state combines hydraulic overlays, landslides, seismic context and ARCUS historical collapses. The bars summarise the territorial signal used to guide recommendations.



06 OPERATIONAL MEANING

PRIMARY FOCUS

Preliminary hydraulic compatibility; Scour susceptibility; Foundation, pier and abutment exposure

For a bridge project context, the dominant driver (Hydraulic) is translated into preliminary attention points. This section does not prescribe design solutions: it frames checks, data requests and comparison with precedents in the province of Torino.

ATTENTION POINTS

- Preliminary hydraulic compatibility
- Scour susceptibility
- Foundation, pier and abutment exposure
- Riverbed dynamics and sediment/debris transport
- Flood/debris scenarios and inspection accessibility

07 PRIORITY ARCUS PRECEDENTS

Priority events are selected by severity, trigger, territorial driver and source reliability. The "Why flagged" column makes explicit why each precedent deserves operational attention.

P1**B00.10.15 - Castellamonte 2000-10-13 / Hydraulic****Total Collapse /
A****P2****B00.10.03 - Ala di Stura 2000-10-13 / Hydraulic****Total Collapse /
A****P3****B00.10.05 - Ala di Stura 2000-10-13 / Hydraulic****Total Collapse /
A**

08 COMPARABLE PRECEDENTS

Comparable precedents share cause, trigger, severity or territorial context. They should be used as historical analogues, not direct predictions.

09 INTERPRETATION

For new Bridge interventions, ARCUS directs first checks toward hydraulic. The reading is derived from historical precedents, hazard overlays and evidence quality: it does not automatically indicate structural criticality, but defines what should be checked next.

10 IMMEDIATE SCREENING ACTIONS

ACTION 1

For new bridge interventions in the province of Torino, prioritise preliminary hydraulic compatibility checks, including flood levels, flow concentration, debris transport and riverbed evolution.

ACTION 2

Treat scour susceptibility as a primary design-screening topic where foundations, abutments or piers interact with active watercourses.

ACTION 3

Compare the planned intervention with documented hydraulic or total-collapse precedents in Castellamonte, Ala di Stura, Felleto, Balme, Chialamberto before defining investigation priorities.

ACTION 4

Use public WMS layers and ARCUS historical precedents as first-level screening, then commission site-specific hydraulic, geotechnical and structural checks before design decisions.

11 DATA TO REQUEST BEFORE DESIGN DECISIONS

DATA 1

Hydraulic model or preliminary hydraulic study

DATA 2

Riverbed survey / bathymetry where relevant

DATA 3

Geotechnical investigation and foundation concept

DATA 4

Historical flood levels

DATA 5

Inspection accessibility constraints

12 RECOMMENDED CLIENT PATH

1. For new bridge interventions in the province of Torino, prioritise preliminary hydraulic compatibility checks, including flood levels, flow concentration, debris transport and riverbed evolution.
2. Treat scour susceptibility as a primary design-screening topic where foundations, abutments or piers interact with active watercourses.
3. Compare the planned intervention with documented hydraulic or total-collapse precedents in Castellamonte, Ala di Stura, Felleto, Balme, Chialamberto before defining investigation priorities.
4. Use public WMS layers and ARCUS historical precedents as first-level screening, then commission site-specific hydraulic, geotechnical and structural checks before design decisions.
5. Where historical precedent density is high, include inspection accessibility and monitoring feasibility in early design discussion.

13 NATIONAL BENCHMARK

Compared with the national ARCUS baseline, the selected province is above average for Triggered-event share, Professional-grade sources, Failure precedent exposure. If triggered-event share is above average, the briefing should be used to anticipate hydraulic, geotechnical and territorial-context checks before site-specific design.

14 DATA COVERAGE & LIMITATIONS

Province-level screening: not design scale.

ARCUS does not certify structural safety or asset condition.

Public overlays guide priorities and must be followed by site-specific checks.

A SOURCE APPENDIX

Main sources linked to events in the selected province, deduplicated by title or URL. Total linked sources in this briefing: 125. The full source table can be exported separately from ARCUS Professional.

**B00.10.15, B00.10.03, B00.10.05... / ARPA
PIEMONTE**
**Evento alluvionale regionale 13-16 ottobre
2000 – monografia ufficiale**
www.arpa.piemonte.it/sites/default/files/media/2023-10...

B00.10.15, B00.10.03, B00.10.05... / ANSA
**10 anni fa alluvione Nord-Ovest. fu inferno
d'acqua – Piemonte e Valle d'Aosta**
www.ansa.it/web/notizie/photostory/primopiano...

**B00.10.15, B00.10.03, B00.10.05... /
JOURNAL PAPER**
**Bridge collapses in Italy across the 21st
century: survey and statistical analysis**
doi.org/10.1080/15732479.2025.2483500

B05.11.01 / CORRIERE DELLA SERA
Bloccata la linea ferroviaria Torino-Roma
[www.corriere.it/Primo_Piano/Cronache/2005/11 Nove...](http://www.corriere.it/Primo_Piano/Cronache/2005/11_Nove...)

B05.11.01 / GAZZETTA D'ASTI
**Crollo della passerella a Villanova: condanne
in Tribunale**
www.gazzettadasti.it/2011/06/22/crollo-della-passerell...

B08.05.02 / REGIONE PIEMONTE
**Evento alluvionale 29-30 maggio 2008 –
relazione tecnica ufficiale**
[www.regione.piemonte.it/web/sites/default/files/media/do...
C...](http://www.regione.piemonte.it/web/sites/default/files/media/do...)

B08.05.02 / METEONETWORK
**Alluvione maggio 2008 in Provincia di
Cuneo: resoconto completo**
www.meteonetwork.it/wp-content/uploads/2017/03/Alluvi...

B17.06.01 / TORINOTODAY
**Cedimento strutturale: è crollato lo storico
ponte di Strambino**
www.torinotoday.it/cronaca/crollo-ponte-chiusella-st...

15 METHODOLOGY SNAPSHOT

ARCUS combines historical collapse evidence, source reliability, spatial relevance, severity, triggers, cause similarity, territorial hazard context and public WMS overlays. The briefing supports preliminary screening and prioritisation: it is not a structural safety certification.

COMPONENT	USE IN THE BRIEFING
Historical evidence	Documented ARCUS records, causes, triggers and severity.
Source reliability	Number, role and quality of available sources.
Spatial relevance	Selected province and administrative provincial boundary.
Hazard context	Public hydraulic/landslide WMS overlays and seismic context.
Comparable precedents	Cause match, trigger match, severity, territorial context and reliability.

16 SCORE AND CLASS LEGEND

CLASS	INTERPRETATION
A	High reliability: multiple sources, preferably institutional or technical.
B	Medium reliability: documented event with limited coverage or source variety.
C	Limited reliability: local, incomplete or to-be-strengthened evidence.
Critical	Maximum attention: strong historical and/or territorial evidence.
High	High attention: significant evidence but not necessarily complete.
Medium / Low	Moderate or low attention, to be read together with available data.